

Project 4: Preference Elicitation

Human-Centric Artificial Intelligence — Method & Study Design Report

1. Task 1 — Feature Representation

The utility function $U(x) = w^T x$ requires a fixed-dimensional feature vector x for each movie. We extract the following features from the IMDB 5000 dataset:

Feature	Type	Justification
IMDB Score	Numeric (normalised)	Direct measure of overall quality and reception
Duration	Numeric (normalised)	Proxy for viewing commitment; users differ in tolerance
Gross Revenue (log)	Numeric (normalised)	Captures commercial popularity and mainstream appeal
Release Year	Numeric (normalised)	Captures recency preference — some users prefer classics
Genre (21 binary flags)	Binary	Most direct driver of movie preference for most users

Numeric features are normalised to $[0,1]$ using min-max scaling so that they are on a comparable scale with the binary genre indicators. Log-scaling is applied to gross revenue before normalisation because the raw revenue distribution is highly right-skewed (a few blockbusters dominate). Rows with missing values in any of these fields are dropped, leaving approximately 3,800 usable movies.

This feature set was chosen to balance expressiveness with simplicity: genres directly capture taste, while the numeric features capture secondary preferences such as quality threshold (IMDB score), time investment (duration), and cultural currency (popularity and recency). Director and cast features were considered but omitted because they create a very high-dimensional, sparse representation that is poorly suited to a linear utility model with few observations.

2. Task 2 — Extending Bradley-Terry to Rankings

2.1 Standard Bradley-Terry Model (Pairwise)

The Bradley-Terry model defines the probability that item i is preferred over item j as:

$$P(i > j) = \exp(U(i)) / (\exp(U(i)) + \exp(U(j)))$$

where $U(i) = w^T x_i$ is the utility of item i . Given a set of pairwise comparisons, the weight vector w is estimated by maximum likelihood, i.e. by maximising the sum of log-probabilities of the observed outcomes. We use gradient ascent with L2 regularisation to prevent divergence.

2.2 Plackett-Luce Extension (Full Rankings)

To model a full ranking $i_1 > i_2 > \dots > i_n$ of n items, we use the Plackett-Luce model, which is the unique consistent extension of Bradley-Terry to rankings. It is defined as:

$$P(i_1 > i_2 > \dots > i_n) = \prod_{k=1}^{n-1} \exp(U(i_k)) / \sum_{j=k}^n \exp(U(i_j))$$

The intuition is straightforward: at each position k in the ranking, the probability of item i_k being placed next (given that items $i_1, \dots, i_{k-1}$ have already been placed) follows the same ratio rule as Bradley-Terry pairwise comparison, but applied to the remaining unranked items. This is equivalent to saying the user repeatedly selects their favourite from the remaining candidates.

This extension is justified by the Luce choice axiom (independence of irrelevant alternatives): the relative probability of preferring i over j does not depend on which other items are in the choice set. This is a natural and widely-used assumption in preference modelling, and makes the model tractable — the log-likelihood decomposes into a sum over positions, enabling efficient gradient-based MLE.

Note that a ranking of n items provides $C(n,2) = n(n-1)/2$ implicit pairwise comparisons, so Design 2 (ranking 10 items) provides up to 45 comparisons per round, compared to 1 per round in Design 1. This is a key efficiency advantage of the ranking interface.

3. Task 3 — User Study Design

3.1 Research Hypothesis

Primary hypothesis: Design 2 (ranking) produces a more accurate estimate of user preferences (measured by recommendation quality) than Design 1 (pairwise comparison) given the same total number of movies evaluated by the participant.

Secondary hypothesis: Design 1 (pairwise) is perceived as less cognitively demanding and more enjoyable than Design 2 (ranking), even if it yields less information per interaction.

3.2 Study Design

We use a **within-subjects design**: each participant completes both interfaces in counterbalanced order (half start with Design 1, half with Design 2) to control for learning effects and individual taste differences. The two sessions are separated by at least 48 hours to minimise carryover effects.

3.3 Participants and Recruitment

We target **40 participants** (20 per counterbalancing condition), recruited via university mailing lists and online platforms (Prolific). Inclusion criteria: age 18+, native or fluent English speaker, self-reported interest in movies (watches at least one movie per month). Participants are compensated at the local minimum wage rate for their time (estimated 20 minutes total).

3.4 Procedure

Step	Description
1. Consent	Participant reads information sheet and provides informed consent
2. Demographics	Short questionnaire: age, gender, movie-watching frequency
3. Session A	Complete first interface (Design 1 or 2, counterbalanced)
4. Break	At least 48 hours between sessions
5. Session B	Complete second interface (the other design)
6. Post-study survey	NASA-TLX cognitive load scale + satisfaction rating per design
7. Ground truth	Participant rates 20 held-out movies on a 1-5 scale (ground truth)

3.5 Evaluation Metrics

Recommendation quality (primary): Kendall's tau between the ranking of the 20 held-out movies predicted by the fitted w and the participant's ground-truth ratings. Higher tau = better preference recovery.

Cognitive load (secondary): NASA-TLX composite score per design. Lower = less demanding.

User satisfaction (secondary): Single 7-point Likert item 'I found this interface enjoyable to use' per design.

Time per elicited comparison: Total session time divided by the number of implicit pairwise comparisons collected. This measures efficiency.

3.6 Statistical Analysis

Primary hypothesis tested with a paired t-test on Kendall's tau scores across participants (one score per participant per design). We use $\alpha = 0.05$ and report Cohen's d as an effect size measure. A power analysis with $d = 0.5$ and power = 0.8 suggests $N = 34$ participants; our target of 40 provides additional headroom for dropouts. Secondary hypotheses tested with Wilcoxon signed-rank tests (ordinal data).

3.7 Controls and Ethical Considerations

Movie sets shown in each session are disjoint to prevent familiarity effects. Movie selection is uniform random to avoid confounding by popularity. Participants may withdraw at any time without penalty. No personally identifying information is collected beyond demographics. Data is stored encrypted and deleted after publication.

4. Summary

Component	Choice
Feature vector	4 numeric + 21 binary genre flags (25 dimensions total)
Pairwise model	Bradley-Terry with linear utility, MLE via gradient ascent
Ranking model	Plackett-Luce (Luce choice axiom extension of Bradley-Terry)
Study design	Within-subjects, counterbalanced, N=40
Primary metric	Kendall's tau on 20 held-out movies
Secondary metrics	NASA-TLX cognitive load, satisfaction, time efficiency
Statistical test	Paired t-test (primary), Wilcoxon signed-rank (secondary)